Transportation Research Part D 115 (2023) 103609

Contents lists available at ScienceDirect

Transportation Research Part D

journal homepage: www.elsevier.com/locate/trd

Corrigendum

Corrigendum to “Energy consumption models for delivery drones: A comparison and assessment” [Transp. Res. Part D: Transp. Environ. 90 (2021) 102668]

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The authors regret that the following corrections need to be made for this article, and apologize for any inconvenience caused. However, it does not affect the results of the analyses or the conclusions from the original contribution.

The authors mistakenly interpreted the drone used in Kirschstein (2020) to have a body weight of 2 kg. The correct value is a body weight of 12 kg. The following are the corrections for the weight and the associated energy consumption reported for the Kirschstein (2020) drone.

In Section 3.3, the following sentence needs to be corrected: ‘Kirschstein (2020) provides analyses for delivery in an urban region (comparing drones with diesel and electric trucks) with large octocopter drones ($m_1 + m_2 = 12$ kg) that carry a 2.5 kg payload ($m_3 = 2.5$ kg), travel at 22.2 m/s, have a flight radius of 9 km (and the trips include 5 min of hovering), power transfer efficiency $\eta = 0.73$, battery charging efficiency $\eta_c = 0.9$ and $P_{avio} = 100$ J/s (as in D’Andrea (2014)). For this drone, the constants in eq.(20) are: $\kappa = 1.15$, $\kappa_2 = 0.502$ and $\kappa_3 = 0.118$; and the loaded $Epm = 131.5$ J/m’. The value for the large octocopter drones should be $m_1 + m_2 = \mathbf{22}$ kg. The value for loaded *Epm* should be $Epm = \mathbf{225.8}$ J/m.

In Table 3a, the entry in column 6 (heading “$m_1 + m_2$ (kg)”) for Kirschstein (2020) should be **22** instead of 12.

In Table 3b, the entry in column 3 (heading “*Epm* (J/m)”) for Kirschstein (2020) should be **213.9** instead of 127.2, and the entry in column 5 (heading “Body Weight (kg)”) should be **12** instead of 2.

In Fig. 3, the energy consumption per meter of point K-8-L should be **208.9** instead of 117.6, so the correct figure is shown below.

In Section 4.1, the following sentence should be corrected: ‘The next four rows for Kirschstein (2020) show a very small increase (3.4%) in *Epm* from including avionics, a very large increase (129%) due to including strong winds of 45 km/hr, and a substantial increase (70.1%) by including both moderate winds (10 km/hr) and energy use for a complete flight profile’. The correct sentence is: ‘The next four rows for Kirschstein (2020) show a very small increase (**2.4%**) in *Epm* from including avionics, a very large increase (**134%**) due to including strong winds of 45 km/hr, and a **moderate** increase (**22.4%**) by including both moderate winds (10 km/hr) and energy use for a complete flight profile’.

In Table 4, the entry in column 4 (heading “$m_1 + m_2$ (kg)”) for Kirschstein (2020) should be **22** instead of 12; the entry in column 8 (heading “*Epm* (J/m)”) should be **208.9** instead of 127.2 for Baseline, **213.9** instead of 131.5 for Avionics, **489.2** instead of 300.6 for Wind (45 km/h) and avionics, and **255.7** instead of 216.4 for a complete flight profile with wind (10 km/h) and avionics; the entry in column 9 (heading “%*Epm* increase to baseline”) should be **2.4%**, **134%**, and **22.4%**, instead of 3.4%, 129%, and 70.1%, respectively. (Since Table 4 has more than one number to correct, we show the corrected entries for Kirschstein (2020) as below.

The authors would like to apologize for any inconvenience caused.

DOI of original article: https://doi.org/10.1016/j.trd.2020.102668.

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https://doi.org/10.1016/j.trd.2023.103609

Available online 11 January 2023

1361-9209/

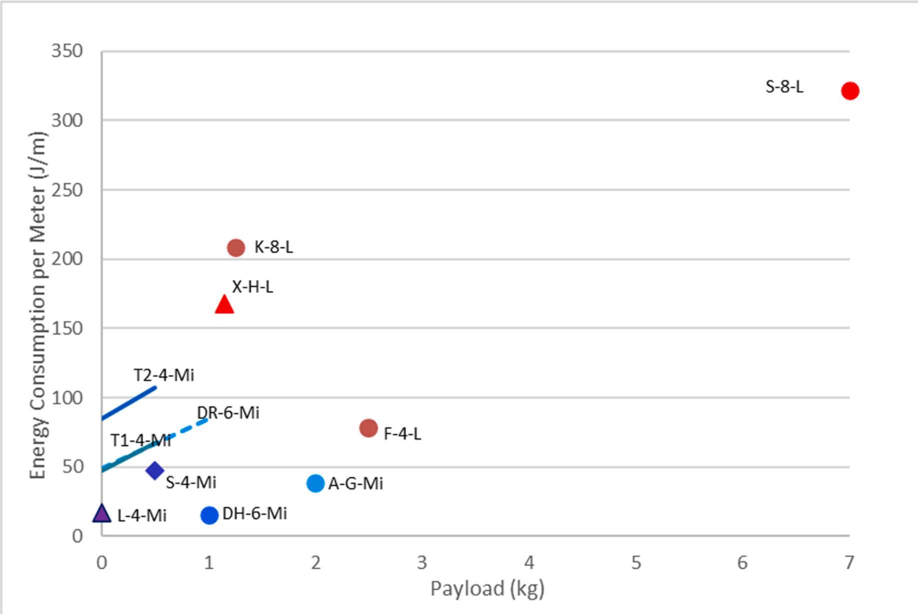

Fig. 3. *Epm* Results from baseline models in Tables 3a and 3b.

Table 4
Key drone energy models with additional conditions.

ID	Reference	Additional conditions to baseline	$m_1 + m_2$ (kg)	Payload[1] (kg)	Airspeed v_a (m/s)	Range[2] (km)	*Epm* (J/m)	%*Epm* increase to baseline
K-8-L	Kirschstein (2020)	Baseline	**22**	1.25[1]	22.2	9	**208.9**	**–**
		Avionics					**213.9**	**2.4%**
		Wind (45 km/h), avionics					**489.2**	**134%**
		A complete flight profile with wind (10 km/h) & avionics					**255.7**	**22.4%**